Teesside University

Emergence of Collective Behaviours: From Human Altruism to AI Safety

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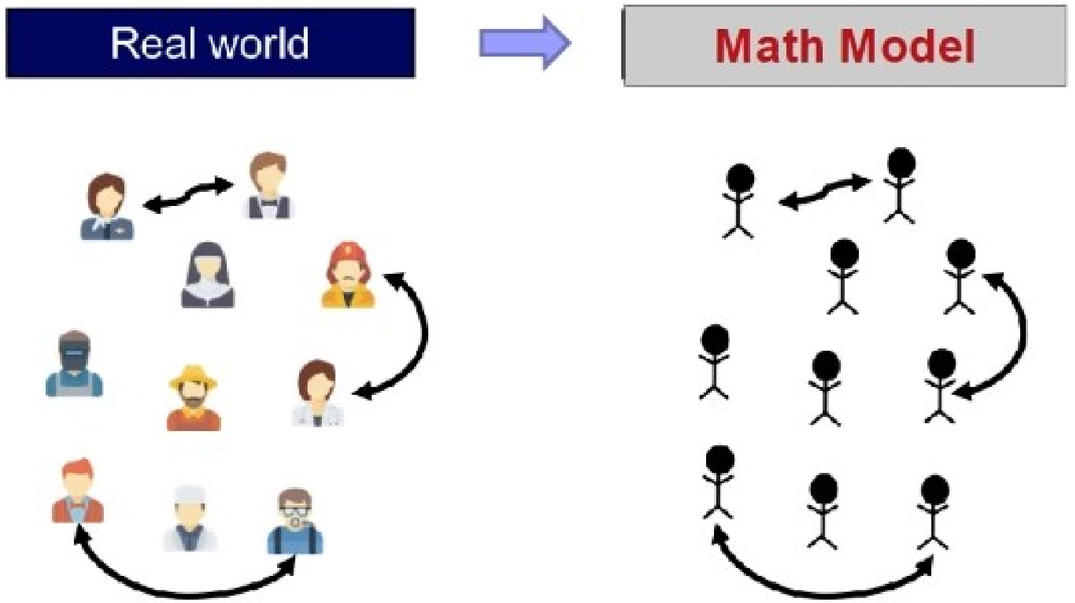

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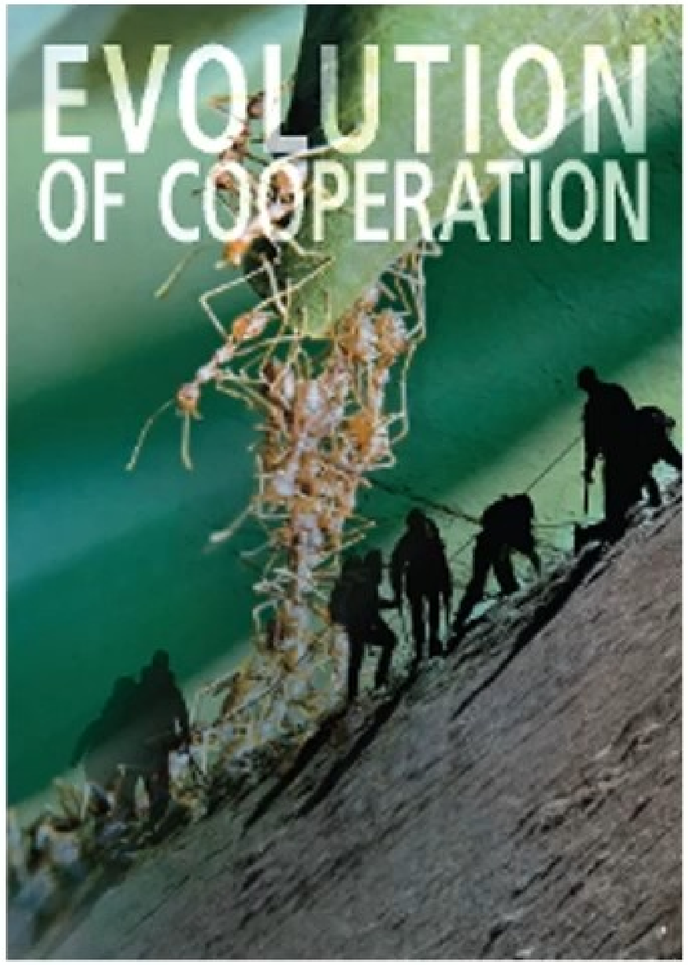
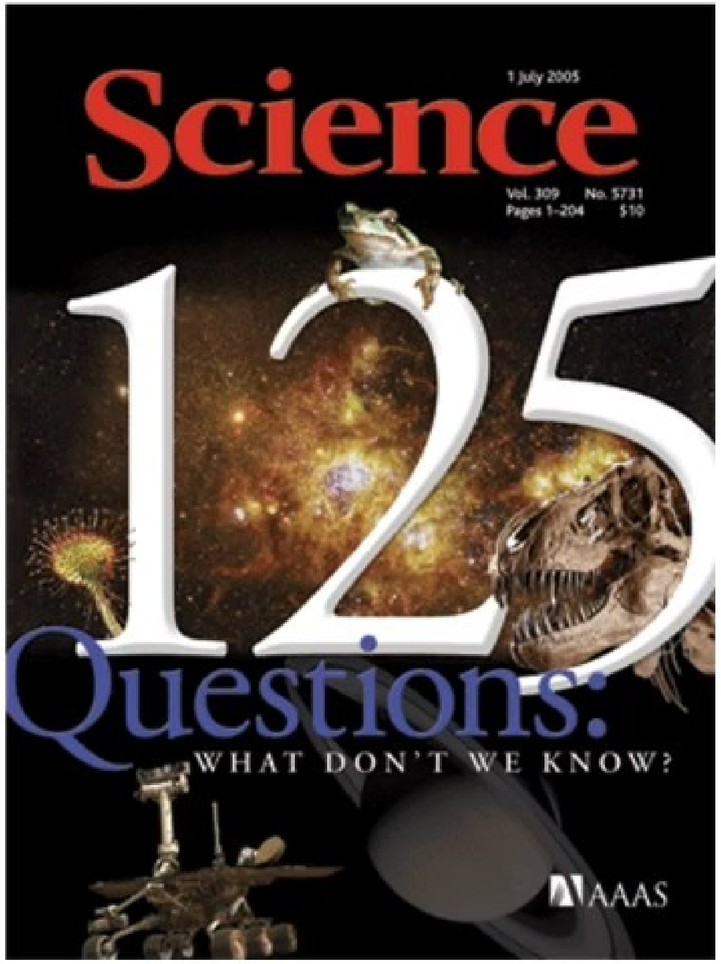
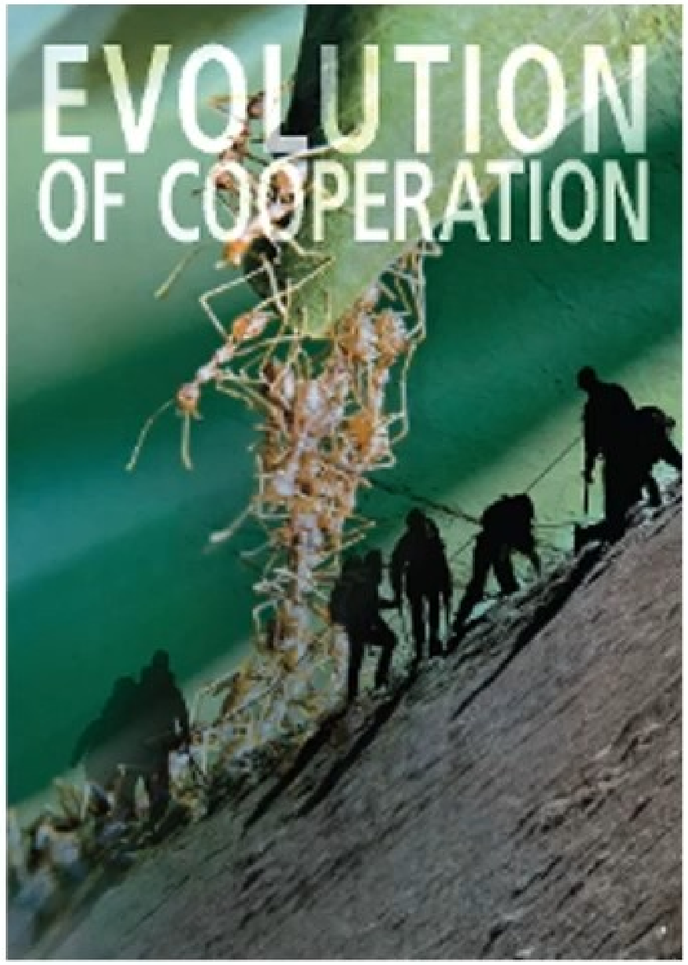
Why we cooperate?

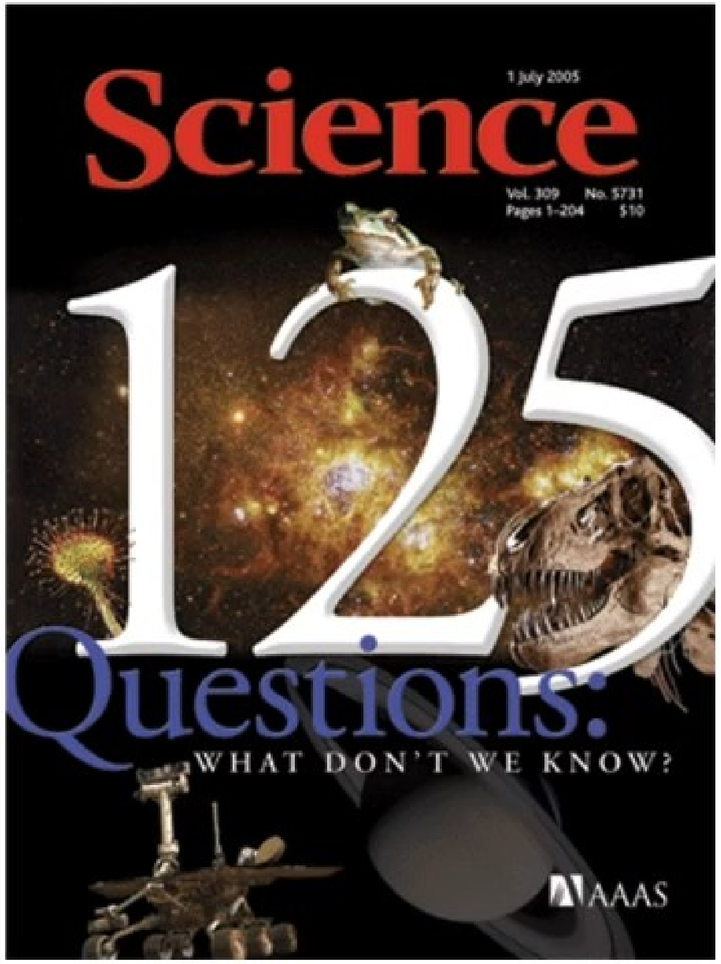




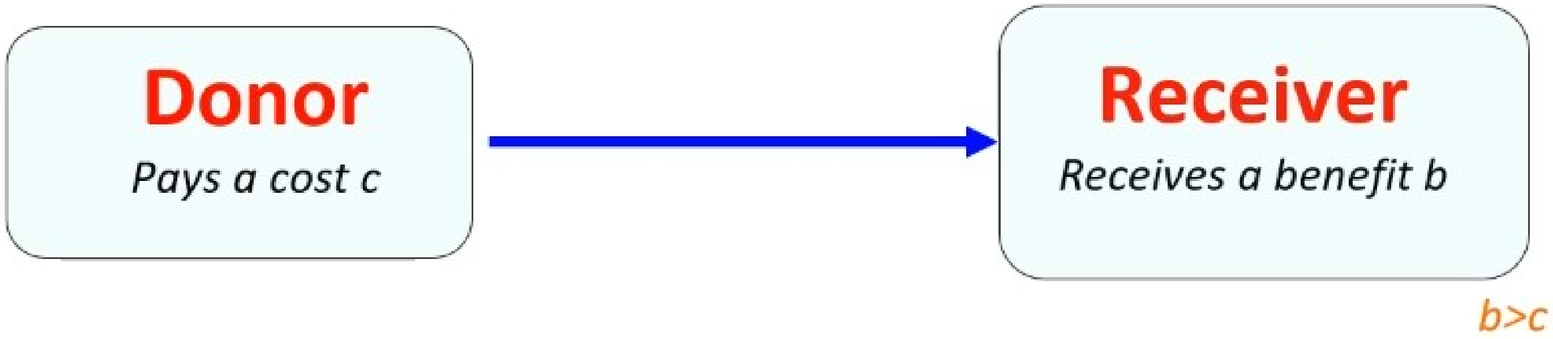
Why we cooperate?

How to engineer pro-sociality?





what is cooperation ?



b>c

what is cooperation ?

Donor

Pays a cost c

→

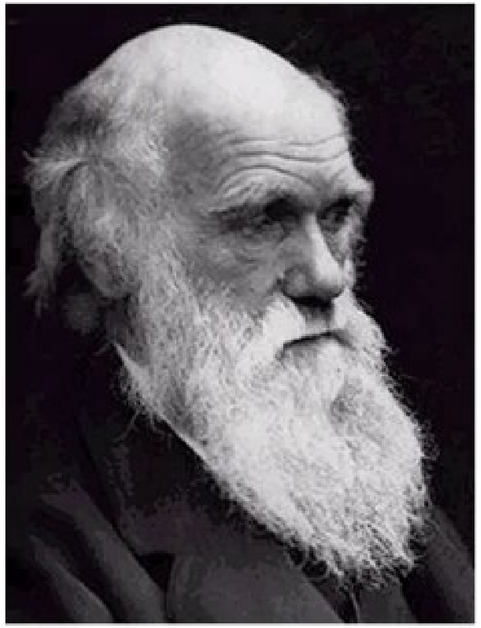
Receiver

Receives a benefit b

b>c

Why should we cooperate?

If natural selection is based on competition, how can it lead to cooperation ?



Collective dilemmas among humans

- collective actions to protect, hunt, nourish, etc.
- water sharing, climate change mitigation
- tax paying and social welfare
- open-source projects

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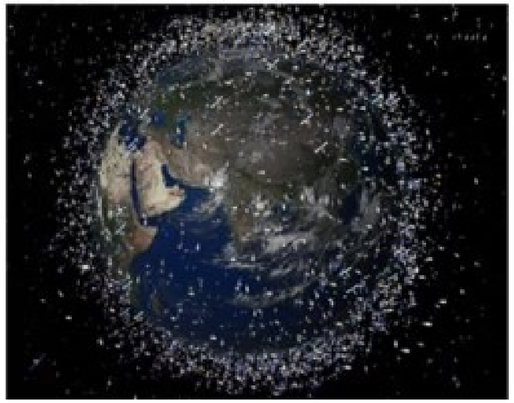
Cooperation and self-organization in engineering sciences

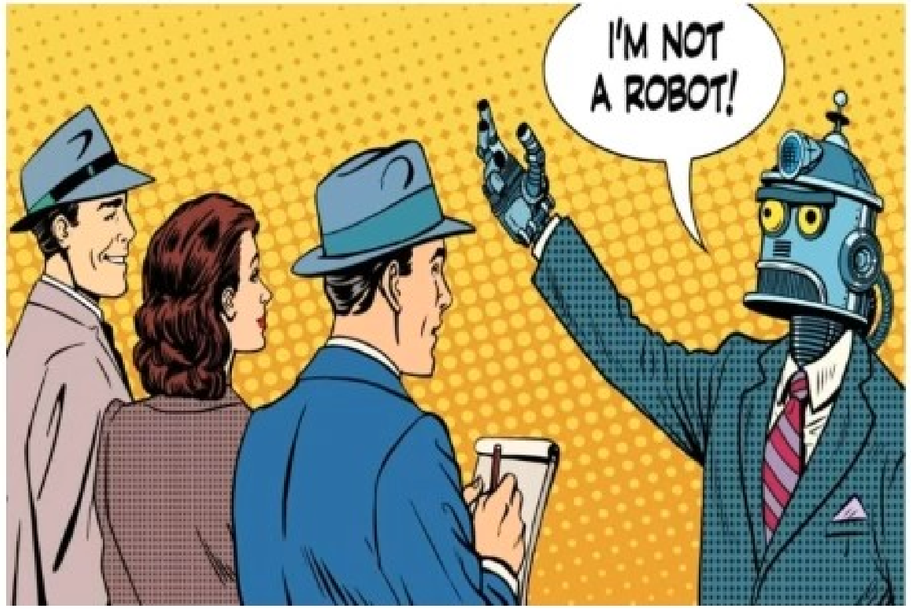


- decentralized control and population-based AI e.g., self-organized task allocation, swarm intelligence, collective robotics, etc.

Rely on

- Understanding and control of the complex nature of self-organization of cooperative behavior.

Cooperation/trust in hybrid human-AI populations



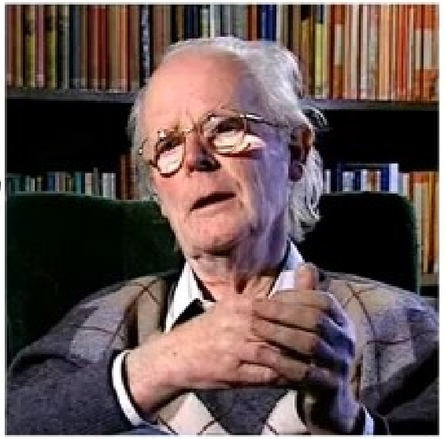
Han, Perret & Powers (2021). **When to (or not to) trust intelligent machines: Insights from an evolutionary game theory analysis of trust in repeated games**. *Cognitive Systems Research*.

Interdisciplinary research

Understanding the **evolution of cooperation** remains a ***fundamental challenge***, for scientists from fields like evolutionary biology, economics, political science, anthropology, computer science, AI, etc.

Many fields… same math!!

John Maynard Smith



Mathematical Framework: Evolutionary Game Theory

Metaphors: Public Goods Game, Prisoner's Dilemma, Ultimatum game, etc.

Video2Any

Slides and notes from video

video2any.com